11. Write a C program for possible keys does the Playfair cipher have? Ignore the fact that some keys might produce identical encryption results. Express your answer as an approximate power of 2.

a. Now take into account the fact that some Playfair keys produce the same encryption results. How many effectively unique keys does the Playfair cipher have?

Aim:

To calculate the number of possible Playfair cipher keys and the approximate number of effectively unique keys.

Procedure:

1. Consider the 25 letters used in the Playfair matrix (I and J are combined).
2. Calculate the total number of keys using \(25!\).
3. Divide by \(5!\times5!\) to account for equivalent row and column arrangements.
4. Display the approximate powers of 2.

Code:

```
#include <stdio.h>
#include <math.h>
int main() {
   int i;
   double total = 1, effective;
   for (i = 1; i <= 25; i++)
      total *= i;
   effective = total / (120.0 * 120.0);
   printf("Total possible keys = %.0f\n", total);
   printf("Approximate power of 2 = 2^%.2f\n",
         log(total) / log(2));
   printf("Effectively unique keys = %.0f\n", effective);
   printf("Effective power of 2 = 2^%.2f\n",
```

```
        log(effective) / log(2));
    return 0;
}
```

Output:

```
Output

Total possible keys = 155112100433309860553031 68
Approximate power of 2 = 2^83.68
Effectively unique keys = 1077167364120207425536
Effective power of 2 = 2^69.87
```

Result:

Thus, the Playfair cipher has approximately \(2^{84}\) possible keys, or \(2^{70}\) effectively unique keys under the row/column-equivalence assumption used here.